

ABHINAV KAUSHIK

Senior AI/ML Engineer

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SUMMARY

Senior AI/ML Engineer with 8+ years delivering production AI solutions across healthcare, telecom, and AIOps. Holds a Master of Science in Machine Learning and AI from Liverpool John Moores University. Experienced in building large-scale forecasting and anomaly-detection systems using Python, SQL, Databricks, and advanced ML models such as LightGBM and Random Forest. Recent work includes a LangGraph analytics platform that cut manual effort by 98% and a production SLA breach predictor that reduced downtime by 67%, demonstrating the ability to create data-driven insights that guide business decisions.

TECHNICAL SKILLS

AI / GenAI: GenAI, LLM, RAG, Agentic AI, LangChain, LangGraph, Claude, OpenAI, Embeddings, Tool Orchestration, Cross-Encoder, Transformers, NLP, Generative AI

Machine Learning: XGBoost, LightGBM, Random Forest, LSTM, SBERT, DBSCAN, Time Series Forecasting, Demand Forecasting, Anomaly Detection, Statistical Modeling, Regression, Deep Learning

Programming / Data: Python, SQL, PySpark, Pandas, NumPy, Flask, Hive, SparkQL, REST API, Pydantic, Unix

MLOps / Cloud: MLflow, Docker, Kubernetes, Jenkins, Bamboo, Bitbucket, DVC, Terraform, Airflow, GCP, Azure, NCI Servers

Search / Observability: Elasticsearch, Kibana, Logging, Log Analysis, Apache Solr, Lucene Indexing, Search and Information Retrieval

Data Platforms: Databricks, Apache Spark, Airflow

Databases: TimescaleDB, MySQL, Vector DB, pgvector, ArangoDB, Cosmos DB

BI / Testing: Power BI, QlikSense, SAP HANA, Selenium

PROFESSIONAL EXPERIENCE

Senior AI Engineer | Optum.AI

Bengaluru, India | Nov 2025 - Present

- Engineered a LangGraph-based analytics system for AARP medical plans, combining stateful multi-agent reasoning with real-time chart generation and reducing manual analysis effort by 98%.
- Enhanced dense retrieval performance by training a cross-encoder for AARP survey data, achieving a 20% increase in retrieval accuracy.
- Built and integrated a lab extraction agent with a LIMS using Claude Opus 4.7 and Amazon Textract, automating work associated with 800 Jira tickets and PRD tasks.

Senior Machine Learning Engineer | Viavi Solutions

Chennai, India | Sep 2024 - Nov 2025

- Deployed a production-grade SLA breach prediction system using Random Forest and LightGBM with FastAPI and MLflow, integrating GPT-4 reasoning for real-time root-cause analysis, resulting in a 67% reduction in downtime.
- Facilitated development of an AI-powered Project Discovery Assistant that analyzes client transcripts and documents with LangChain and LangGraph, achieving a 90% match rate of technical use cases with existing solutions.
- Built 30+ AI application prototypes and mockups for AIOps and GeoPerformance clients, translating requirements into functional POCs for validation and demonstration.

Data Scientist | Viavi Solutions

Bengaluru, India | Nov 2021 - Sep 2024

- Applied Sentence-BERT and DBSCAN to enhance semantic alarm classification, achieving a 73.5% improvement across graph-based AIOps infrastructure.
- Used Isolation Forest and Holt-Winters for anomaly detection and demand forecasting across 500+ telecom network devices with 15+ KPIs, enabling proactive network planning.
- Delivered technical POCs and consulting for 15+ clients across 3+ AIOps and ML platforms, contributing to an 11% increase in POC and mentoring revenue.

Data Analyst | United Health Group

Gurugram, India | Jun 2021 - Nov 2021

- Collaborated with stakeholders on ad-hoc analysis and customer-retention dashboards using Hive, SQL, and Power BI.
- Developed a Python- and Flask-based parser to automate data extraction and transformation, supporting downstream analytics and reporting.

Data Engineer | Nippon Data Systems

Gurugram, India | Nov 2019 - May 2021

- Engineered SQL and PySpark pipelines to automate BAU medical-audit reporting, reducing manual effort by 95%.
- Built real-time observability using Elasticsearch and Kibana, improving traceability efficiency by 80% for the COVID QRT team.
- Developed a Flask-based enterprise search engine with Apache Solr and Lucene indexing, enabling centralized search across organizational data.

Associate Consultant | Saksoft

Gurugram, India | Dec 2017 - Nov 2019

- Conceptualized customer dashboards with QlikSense, SQL, and SAP HANA, increasing user engagement by 40%.
- Developed an XGBoost propensity model to predict personal loan initiation within 7 days, enabling targeted outreach.
- Automated performance testing with Selenium and Python, reducing testing effort by 90%.

PROJECTS

Think Tree - ML Study Platform | Live Platform | <https://think-tree-ai.vercel.app/>

- Built a non-profit AI-powered ML study platform for people with ADHD, helping users organize learning and maintain study context.

EDUCATION

Master of Science, Machine Learning and AI | Liverpool John Moores University

Jan 2020 - Jan 2022

Bachelor of Technology, Information Technology | Jamia Hamdard

Aug 2013 - Aug 2017

ADDITIONAL INFORMATION

- German: A1 (in progress)
- Network Intern, Airport Authority of India
- Hadoop Intern, Ducat Technologies